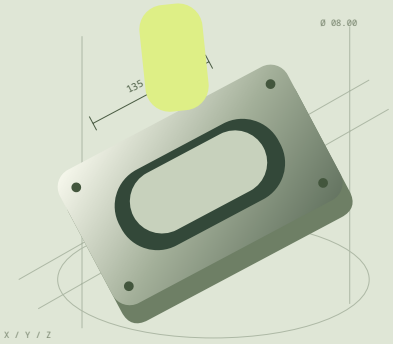
PROFESSIONAL PORTFOLIO · CÁRTAMA, MÁLAGA

Ideas that take shape.

Nicolás Acebal Colli

Industrial design engineering and product development.

Founder and technical lead at Acebal Instrumentos, with three years of experience in aeronautical training systems. I connect design, manufacturing and teaching, and am expanding into web development through my Product Engineering master's studies at Evolve.



Conceptual graphic study of form and function.

Industrial design | **Manufacturing** | **Educational products** | **Web development**

02 / Design & manufacturing

03 / Educational products

04 / Digital development

05 / Systems & ongoing projects

06 / Background & contact

nacecol01@gmail.com · Cártama, Málaga

01 / DESIGN & MANUFACTURING

From design to manufacture.

Professional experience in products, engineering and technical documentation.



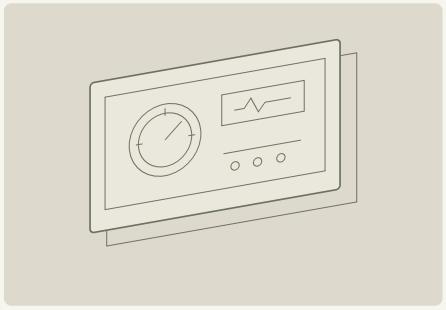
Concept illustration; not a photograph of the project.

Founder & technical lead · Since 2024

Acebal Instrumentos

Design and manufacture of musical instruments. Development from concept through manufacturing, including production, suppliers and quality management.

> Outcome: musical instruments designed and manufactured through my own venture.



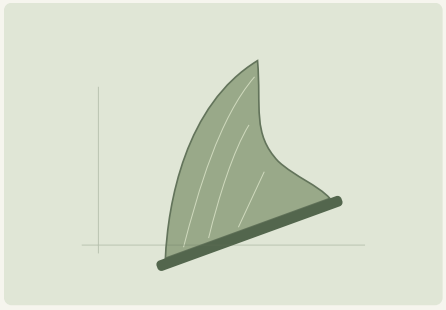
Concept illustration; not a photograph of the project.

Design & development engineering · 2022–2025

Aerodynamics Academy

Development of systems for aeronautical technician training: CAD design, technical documentation and project coordination.

> Contribution: technical design and system documentation for aeronautical training.



Concept illustration; not a photograph of the project.

Product design · 3D modelling

Surf components

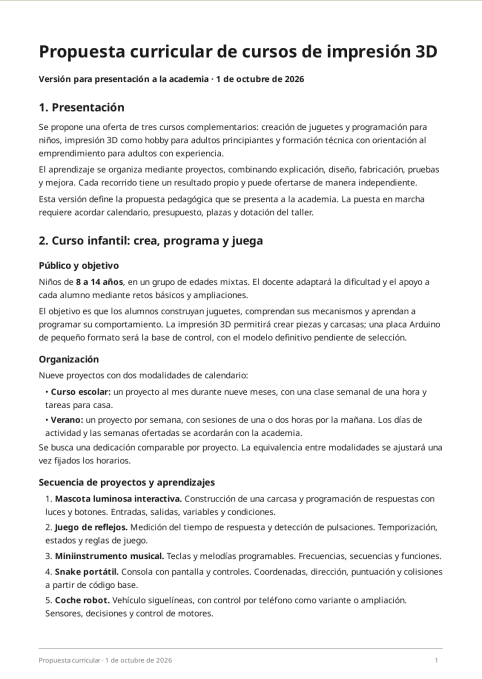
Design and development of surfboard parts, connecting technical modelling with manufacturing.

> Contribution: component development and product-focused modelling.

02 / EDUCATIONAL PRODUCTS

Create to teach.

Original course design, educational product development and active teaching.



Propuesta curricular de cursos de impresión 3D

Versión para presentación a la academia · 1 de octubre de 2026

1. Presentación

Se propone una oferta de tres cursos complementarios: creación de juguetes y programación para niños, impresión 3D como hobby para adultos principiantes y formación técnica con orientación al emprendimiento para adultos con experiencia.

El aprendizaje se organiza mediante proyectos, combinando explicación, diseño, fabricación, pruebas y mejora. Cada recorrido tiene un resultado propio y puede ofertarse de manera independiente.

Esta versión define la propuesta pedagógica que se presenta a la academia. La puesta en marcha requiere acordar calendario, presupuesto, plazas y dotación del taller.

2. Curso infantil: crea, programa y juega

Público y objetivo

Niños de **8 a 14 años**, en un grupo de edades mixtas. El docente adaptará la dificultad y el apoyo a cada alumno mediante retos básicos y ampliaciones.

El objetivo es que los alumnos construyan juguetes, comprendan sus mecanismos y aprendan a programar su comportamiento. La impresión 3D permitirá crear piezas y carcasas; una placa Arduino de pequeño formato será la base de control, con el modelo definitivo pendiente de selección.

Organización

Nueve proyectos con dos modalidades de calendario:

- **Curso escolar:** un proyecto al mes durante nueve meses, con una clase semanal de una hora y tareas para casa.
- **Verano:** un proyecto por semana, con sesiones de una o dos horas por la mañana. Los días de actividad y las semanas ofertadas se acordarán con la academia.

Se busca una dedicación comparable por proyecto. La equivalencia entre modalidades se ajustará una vez fijados los horarios.

Secuencia de proyectos y aprendizajes

1. **Mascota luminosa interactiva.** Construcción de una carcasa y programación de respuestas con luces y botones. Entradas, salidas, variables y condiciones.
2. **Juego de reflejos.** Medición del tiempo de respuesta y detección de pulsaciones. Temporización, estados y reglas de juego.
3. **Mininstrumento musical.** Teclas y melodías programables. Frecuencias, secuencias y funciones.
4. **Snake portátil.** Consola con pantalla y controles. Coordenadas, dirección, puntuación y colisiones a partir de código base.
5. **Coche robot.** Vehículo siguelíneas, con control por teléfono como variante o ampliación. Sensores, decisiones y control de motores.

Propuesta curricular · 1 de octubre de 2026 1

Actual document: 3D printing curriculum proposal, October 2026.

3D printing

Introductory and specialist courses with pathways for children, adult beginners and experienced adults.

Physical products + applications

For the children’s course, I have developed both the products and the applications that make them work, bringing together design, manufacturing and programming.

Three defined learning pathways

The curriculum organises children’s training, adult introductory training and advanced training. This is a curriculum-design outcome, not a count of delivered courses.

Original guitar courses

Creation of courses that I am beginning to sell, connecting my musical experience with an educational offering of my own.

Currently teaching

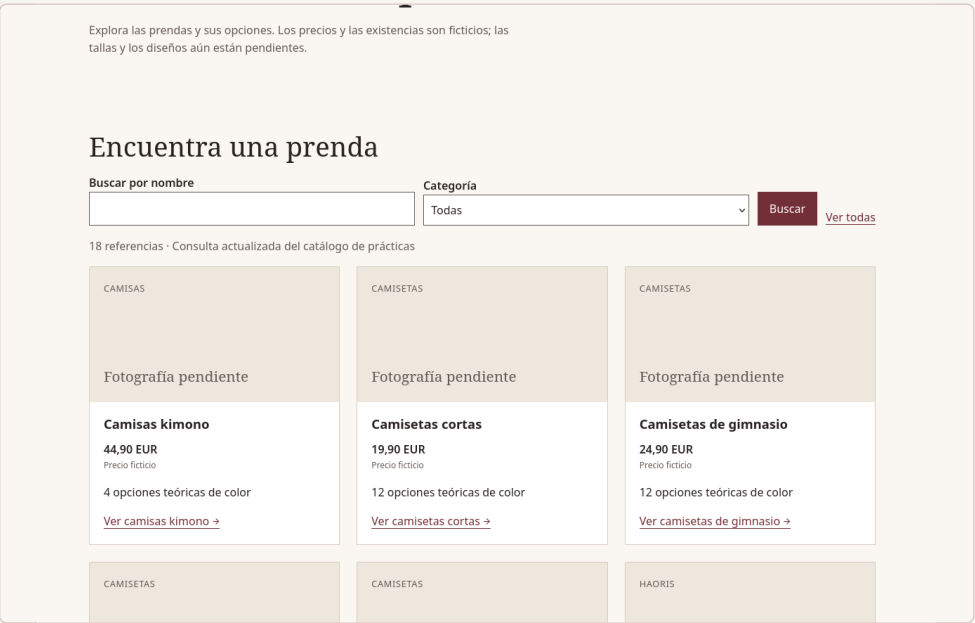
I currently teach. My experience includes supporting engineering students in design, 3D modelling and technical problem-solving.

03 / DIGITAL DEVELOPMENT

¡Kimono, bambino!

A practice shop connecting Python business logic with a Django web interface.

Python | Django | SQLite | HTML / CSS | Git



Actual desktop screenshot of the practice catalogue (Spanish interface).



The same application on mobile.

What I have developed

- Catalogue with search and filters.
- Persistent shopping cart per account.
- Simulated orders and inventory management.

How it works

Python modules shared by command-line and web interfaces, SQLite persistence, and forms with authentication and CSRF protection.

Available result

A working demo with 18 catalogue entries and a practice flow through catalogue, cart and order. Prices, stock and payments are simulated; this is not a commercial store.

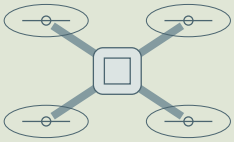
04 / IT & SYSTEMS

Technology for getting things done.

Advanced computer skills, experience with personal systems and training in project information management.

Personal infrastructure	NAS, Linux and self-hosted services with Podman. Private remote access through Tailscale and HTTPS.
Notes & knowledge	Obsidian, LiveSync and CouchDB to organise projects and synchronise notes across devices, with encryption and access permissions.
Cybersecurity	Fundamentos y estrategias avanzadas en Ciberseguridad · VIRENSIS · Generación Digital Pymes. 150 hours, completed 24 June 2026, result: Pass.
Project information	Two completed SolidWorks PDM courses and extensive training in information and technical documentation management.

Ongoing projects



Drone development

Team established; awaiting the first prototypes.



Cleaning robot

Exploring the conversion of a Parkside PWD 25 C4 into an autonomous system. Design and feasibility stage.

Concept diagrams; they do not depict built prototypes.

05 / BACKGROUND & CONTACT

Engineering thinking. A drive to make.

Since 2024

Acebal Instrumentos

Founder and technical lead: instrument design, product development and manufacturing management.

2022–2025

Aerodynamics Academy

Design and development engineering for aeronautical training systems, documentation and coordination.

Since 2021

Teaching & mentoring

Current teaching activity and support with design, 3D modelling and technical problem-solving.

Education

Industrial Design Engineering and Product Development · University of Málaga. One exchange year in Skövde, Sweden.

Master’s studies in Product Engineering · Evolve · In progress.

Additional training: cybersecurity (150 h, VIRENSIS) and two SolidWorks PDM courses.

SolidWorks | AutoCAD | Rhino 3D | Blender | SolidWorks PDM | Python | Django | Git

Let’s bring it to life.

Get in touch to discuss a project, a collaboration or a professional opportunity.

nacecol01@gmail.com

Cártama, Málaga

nicolas-acebal-colli.gardenpond8.chatgpt.site

Web portfolio: access depends on the site’s sharing permissions.